Kovalyshyn Oleh - Assignment 2

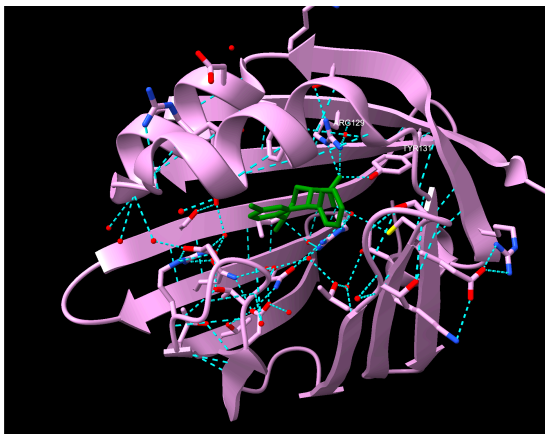
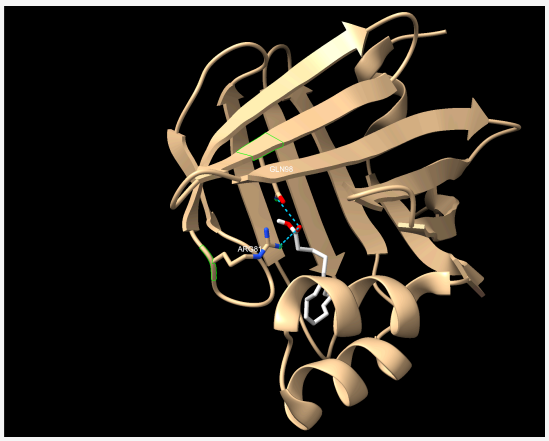
Part I: Target and Ligand pre-processing

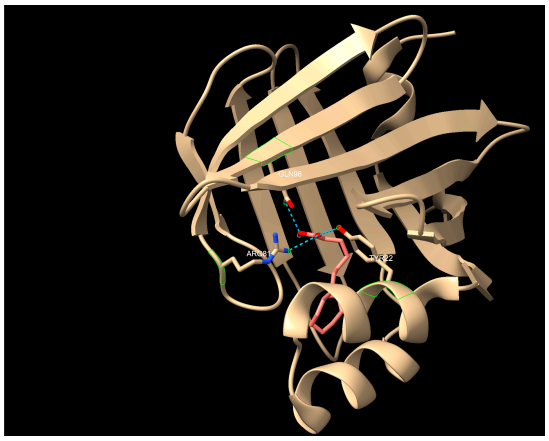
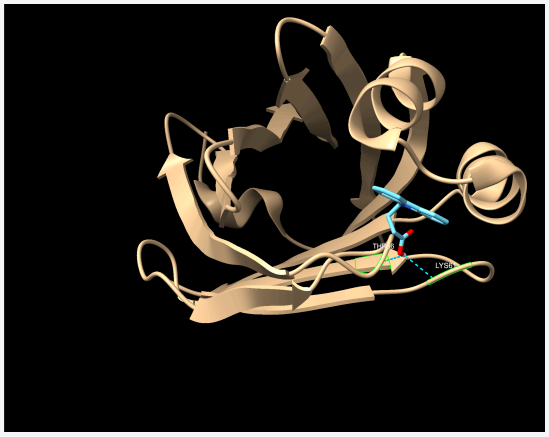
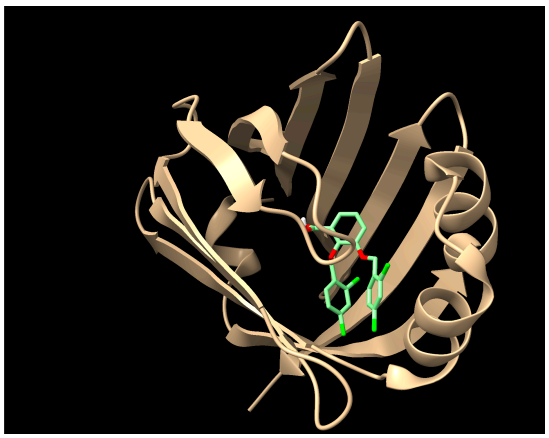
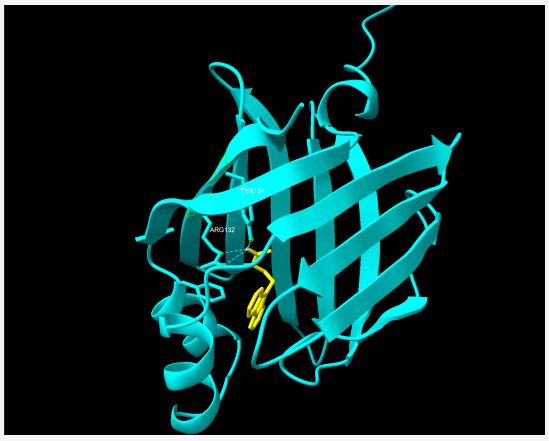
Target: FABP5 (UniProt), 135AA, beta-barrel protein.
Transports fatty acids and lipids, endocannabinoids, regulates methabolism.

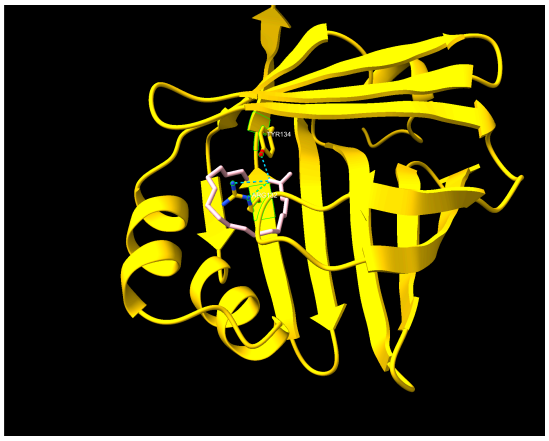
Natural ligands: PALMITIC ACID, OLEIC ACID
Artificial ligands: CHEMBL247298 (Compound 13), CHEMBL185595 (carbazole butanoic acid)

Structure: 7G0B (X-RAY DIFFRACTION, Resolution: 1.47 Å)

I decided to compare wild-type (WT) structure with 4 other Vina-bound (AD) ligands and 2 AlphaFold3 bound ligands (AF) (the other two had no CCD so AF3 couldn't bind them). Wild type is 7-bromo-1-methyl-5-phenyl-2,3,4,5-tetrahydro-1-benzazepine-4-carboxylic acid, the original molecule proven to bind FABP5 with X-ray.

Experiment	Pose	Binding pocket	Key residues	PRODIGY -LIG ΔG
WT		Internal cavity	ARG 129, TYR 131, Internal HOH	-10.08*
AD - PAL			ARG 81, GLN 98	-5.44

Experiment	Pose	Binding pocket	Key residues	PRODIGY -LIG ΔG
AD - OLA			TYR 22, ARG 81, GLN 98	-5.47
AD - C13			THR 56 LYS 61	-5.83
AD - CRZ			No H-bonds found	-5.15
AF - CRZ			ARG 132, TYR 134	-8.84

Experiment	Pose	Binding pocket	Key residues	PRODIGY -LIG ΔG
AF - OLA			ARG 132, TYR 134	-9.52

* - the structure's own card report IC50=2.3 μM -> affinity of ~7.7 kcal/mol. Possible error of ~2 kcal/mol.

Several observations can be made from this. First, Autodock Vina constantly overestimates binding affinity, compared to Prodigy. For example, for Compound 13, Vina estimated affinity at -9.16 kcal/mol, but Prodigy reported only -5.83 kcal/mol (and from the WT we expect this to be an overshot). Second, Vina is inconsistent when it comes to interacting residues. I looked mainly at H-bonds and an argument can be made that in case of enclosed ligand like this one, main type of interaction would be Van der Waals forces. But AF3 co-folds don't allow to dismiss H-bonds so easily. In both experiments, AF3 provided identical H-bonds, and all of them are much closer to WT interactions than those predicted by Vina. After all, Vina failed to produce H-bonds for carbazole acid.

Another thing is water. In case of FABP5, ligand is not only held in place by H-bonds and Van der Waals. There's a body of evidence that water molecules, encapsulated in the cavity are in fact ordered and play a critical role in stabilizing the ligand (V. A. Likić et al., 2000, Mihajlovic & Lazaridis, 2007). We can indeed see that in WT. A water molecule forms an H-bond with the ligand and, simultaneously, with other amino acids. This stabilization was not accounted for in binding models, neither in Vina, nor in AF3. Including water would probably increase resulting binding affinity.

Overall, AF3 solved the task better. Both AF3 and Vina correctly placed molecules in roughly the same spot, but it's worth mentioning that Vina was nudged into it by providing bounding box. On the other hand, beta barrels are quite popular type of proteins, so it's no surprise it was no challenge for AF3.